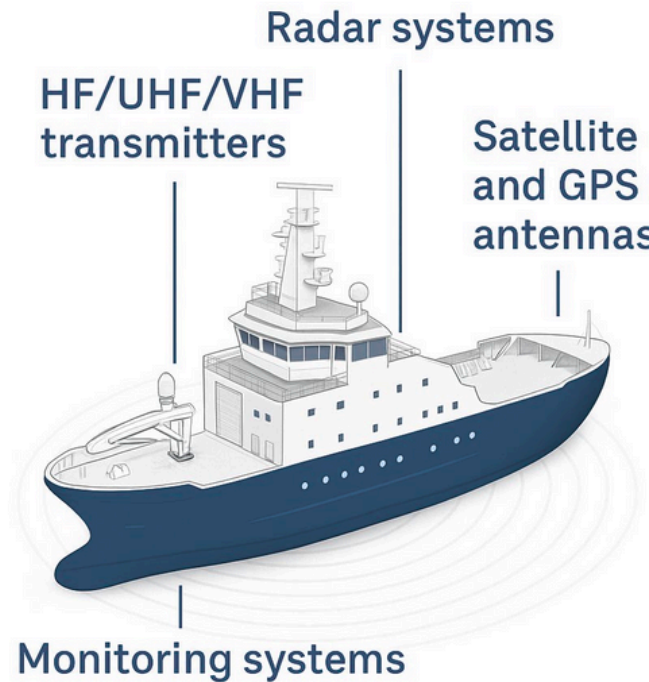


## EMC & EMI in Ship Windows – Importance & Our Capabilities

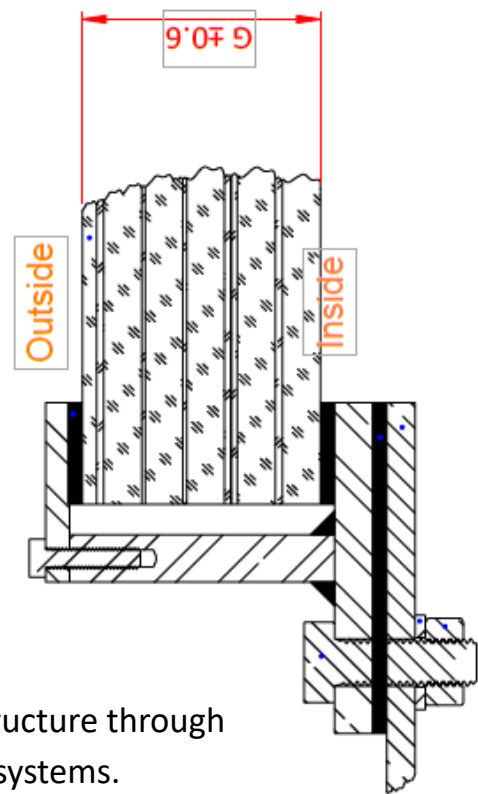
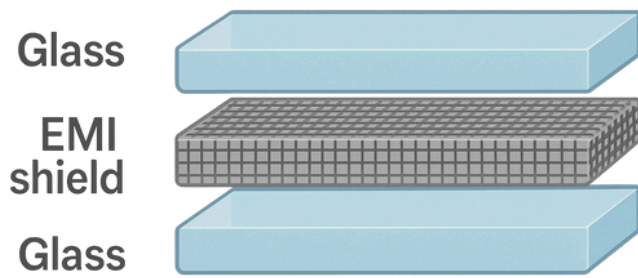
Electromagnetic Compatibility (EMC) refers to the ability of electrical and electronic systems to function properly in their electromagnetic environment without introducing intolerable electromagnetic disturbances to anything in that environment.

Electromagnetic Interference (EMI), on the other hand, is the unwanted disturbance caused by electromagnetic radiation or conduction which can affect the performance of onboard electronic equipment, such as navigation, communication, radar, or control systems.



As a window manufacturer, our responsibility extends beyond mechanical strength and watertightness. Our windows, particularly those made with shielded or laminated glass constructions, must prevent electromagnetic leakage or penetration which can compromise system reliability or security.

We have experience with EMI-tested marine windows in the frequency range of 2 GHz to 18 GHz. This covers a wide range of maritime communication and radar systems. Shielding effectiveness has been verified to reduce EM signal transmission, offering protection from EMI-related malfunctions.



This glass combination is integrated into a robust structure through Karaca's bolted or welded frame installation systems.

Furthermore, as Karaca, we are able to design and manufacture this glass with additional features such as heating and compliance with bullet-resistant classifications.

The use of EMI/EMC shielded glazing ensures:

- Enhanced protection of navigation and control systems,
- Compliance with shipowner and naval authority specifications,
- Seamless integration with Karaca's welded or bolted window systems

These products are developed to comply with relevant international standards, including but not limited to:

- MIL-STD-461: U.S. military standard specifying control of EMI characteristics of equipment.

# Testing

| Frekanslar (MHz)<br><i>Frequency (MHz)</i> | Ekranlama Etkinliđi (dB)<br><i>Attenuation (dB)</i> |
|--------------------------------------------|-----------------------------------------------------|
| 2000                                       | 51.5                                                |
| 3000                                       | 49.86                                               |
| 5000                                       | 47.6                                                |
| 8000                                       | 60.38                                               |
| 10000                                      | 59.4                                                |
| 12000                                      | 54                                                  |
| 16000                                      | 62.8                                                |
| 18000                                      | 59.65                                               |

\*Ekranlama Etkinliđi deđerlerinin aritmetik ortalaması 55.07dB dir. /  
*The arithmetic average of the Shielding Efficiency attenuation is 55.07dB*

